

Weyl Symmetry of the Refined 3D Index for $m004$

Manual Verification via g_{NZ} Matrix and Affine Shift

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1 Overview

The manifold $m004$ is the figure-eight knot complement, cusped hyperbolic with $n = 2$ ideal tetrahedra and $r = 1$ cusp. All data below are extracted from the SnapPy triangulation by the `manifold_index` codebase and verified numerically.

Goal. Show that the refined 3D index

$$\mathcal{I}(m, e) \in \mathbb{Z}[\eta^{\pm 1/2}][[q^{1/2}]]$$

satisfies the *Weyl symmetry*: defining

$$f(\eta) := \eta^{bm+ae} \mathcal{I}(m, e), \quad (1)$$

the function f is palindromic, $f(\eta) = f(\eta^{-1})$, with $a = \frac{1}{2}$ (half-integer) and $b = -\frac{1}{2}$ (half-integer).

Remark 1. Because $a = \frac{1}{2} \notin \mathbb{Z}$, the corresponding “subtract a , add $2b$ ” row prescription on g_{NZ} produces a matrix T with a *half-integer* entry ($-a = -\frac{1}{2}$), which is not a valid lattice basis change ($T \notin \text{GL}(n, \mathbb{Z})$). Hence the prescription is **not Dehn-filling compatible**: Dehn filling requires all charge-lattice operations to be integral.

We do this by:

1. Writing down g_{NZ} and g_{NZ}^{-1} explicitly.
2. Identifying the charge vector κ and the hard-edge coupling H_{arg} .
3. Deriving the affine centre formula and extracting a, b from pure-meridian and pure-longitude pairs.
4. Verifying the palindrome property (1) term by term for several explicit (m, e) values.

2 Neumann-Zagier Data

2.1 The g_{NZ} matrix

The Neumann-Zagier matrix $g_{\text{NZ}} \in \text{Sp}(2n, \mathbb{Z})$ with $n = 2$ encodes how external charges (meridian M , hard edge H , longitude/2 $L/2$, conjugate Γ) couple to the $2n = 4$ logarithmic tet variables. Rows are ordered $(M, H, L/2, \Gamma)$; columns are the four log-tet variables.

$$g_{\text{NZ}} = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & -1 & 2 \\ 0 & 1 & 0 & 2 \\ 1 & 0 & 0 & 0 \end{pmatrix} \quad (\text{rows: } M, H, L/2, \Gamma) \quad (2)$$

Symplecticity check: $g_{\text{NZ}}^T J_{2n} g_{\text{NZ}} = J_{2n}$ where $J_{2n} = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$. `is_symplectic = True.`
✓

2.2 The inverse g_{NZ}^{-1}

Using the symplectic identity $g_{\text{NZ}}^{-1} = J_{2n}^{-1} g_{\text{NZ}}^T J_{2n}$, one obtains exact integer entries:

$$g_{\text{NZ}}^{-1} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 2 & 0 & -1 & -2 \\ 0 & -1 & 1 & 1 \\ -1 & 0 & 1 & 1 \end{pmatrix} \quad (\text{rows: } M, H, L/2, \Gamma) \quad (3)$$

Verified: $g_{\text{NZ}}^{-1} \cdot g_{\text{NZ}} = I_4$. ✓

3 Charge Vector and Hard-Edge Coupling

3.1 The κ vector

The code `build_kappa` constructs, for $n = 2$, $r = 1$, one internal edge $n_{\text{int}} = n - r = 1$:

$$\kappa = \underbrace{(m, 0)}_{\text{position: ext, int}} \mid \underbrace{(e, e_{\text{int}})}_{\text{momentum: ext, int}} = (m, 0, e, e_{\text{int}}),$$

where:

- $m \in \mathbb{Z}$ is the meridian charge (external, fixed),
- 0 is the internal position charge (always zero by construction),
- $e \in \frac{1}{2}\mathbb{Z}$ is the longitude/2 charge (external, fixed),
- $e_{\text{int}} \in \frac{1}{2}\mathbb{Z}$ is the summation variable.

3.2 Hard-edge coupling H_{arg}

The hard-edge tet argument is the second entry of $g_{\text{NZ}}^{-1}\kappa$:

$$H_{\text{arg}} = (g_{\text{NZ}}^{-1})_{\text{row } H} \cdot \kappa = (2, 0, -1, -2) \cdot (m, 0, e, e_{\text{int}}) = 2m - e - 2e_{\text{int}}.$$

This is a linear function of the summation variable e_{int} .

3.3 Relation to the "subtract a , add $2b$ " prescription

The prescription "subtract a units of hard edge from the meridian row, and add $2b$ units to the longitude/2 row of g_{NZ} " corresponds to the row-shearing matrix

$$T(a, b) = \begin{pmatrix} 1 & -a & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 2b & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Convention A ($a = -1$, $b = +\frac{1}{2}$, integer a):

$$T_A = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \in \text{GL}(4, \mathbb{Z}). \quad \text{Lattice-compatible. } \checkmark$$

Convention B ($a = +\frac{1}{2}$, $b = -\frac{1}{2}$, half-integer a):

$$T_B = \begin{pmatrix} 1 & -\frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \notin \text{GL}(4, \mathbb{Z}). \quad \text{Not lattice-compatible. } \times$$

Since the internal position charge is *identically zero* ($\kappa[1] = 0$ by `build_kappa`), one has $T\kappa = \kappa$ for *both* conventions, and therefore

$$z^{\text{new}} = g_{\text{NZ}}^{-1} T^{-1} T \kappa = g_{\text{NZ}}^{-1} \kappa = z^{\text{old}}.$$

Either row transformation has no effect on the refined index. The Weyl symmetry is instead revealed through an affine shift of the summation variable e_{int} , as described next.

4 Affine-Shift Weyl Symmetry

4.1 Centre formula

Define the *Weyl multiplier* as the exponent in (1):

$$\mu(m, e) := b m + a e = -\frac{1}{2} m + \frac{1}{2} e = \frac{e-m}{2}. \quad (4)$$

So $f(\eta) = \eta^{(e-m)/2} \mathcal{I}(m, e)$ is palindromic. Equivalently, the *centre* of the η -distribution in $\mathcal{I}$ is

$$c(m, e) = -\mu(m, e) = \frac{m-e}{2}, \quad (5)$$

meaning $\mathcal{I}(m, e) = \eta^{c(m, e)} \cdot f(\eta)$ with f palindromic.

The Weyl symmetry is the *palindrome of the full sum* over e_{int} after extracting η^c . Explicitly,

$$\mathcal{I}(m, e) = \sum_{e_{\text{int}}} \eta^{e_{\text{int}}} \prod_{\Delta} I_{\Delta}(z_{\Delta}(m, e, e_{\text{int}})), \quad (6)$$

and the claim is that the coefficient of η^{c+k} equals the coefficient of η^{c-k} for all k and at every order in q . This follows from the symmetry $I_{\Delta}(-x, -p) = I_{\Delta}(x, p)$ of the quantum dilogarithm: under $e_{\text{int}} \rightarrow 2c - e_{\text{int}}$ the full tet-argument vector maps to its negative, while $\eta^{e_{\text{int}}} \rightarrow \eta^{2c} \cdot \eta^{-e_{\text{int}}}$. Hence the sum is invariant under $\eta \rightarrow \eta^{-1}$ after extracting η^c .

4.2 Extraction of a and b from pure pairs

The parameters a and b in $\mu(m, e) = b m + a e$ are extracted from the *leading- q distribution* of the refined index at pure-meridian and pure-longitude charges by reading off the centre $c = -\mu$:

Meridian pairs $(\pm m_0, 0)$:

$$b = \frac{\mu(+m_0, 0)}{m_0} = -\frac{c(+m_0, 0)}{m_0}.$$

Numerically for $m_0 = 2$: centre of $\mathcal{I}(+2, 0)$ is $c = +1$, centre of $\mathcal{I}(-2, 0)$ is $c = -1$. Hence $b = -1/m_0 \rightarrow \mathbf{b} = -\frac{1}{2}$.

Longitude pairs $(0, \pm e_0)$:

$$a = \frac{\mu(0, +e_0)}{e_0} = -\frac{c(0, +e_0)}{e_0}.$$

Numerically for $e_0 = 1$: centre of $\mathcal{I}(0, -1)$ is $c = +\frac{1}{2}$, centre of $\mathcal{I}(0, +1)$ is $c = -\frac{1}{2}$. Hence $a = -(-\frac{1}{2})/1 \rightarrow \mathbf{a} = +\frac{1}{2}$.

Validity: $b = -\frac{1}{2}$ (half-integer ✓), $a = +\frac{1}{2}$ (half-integer; *not* an integer — see remark below), no warnings from `compute_ab_vectors`.

Remark 2 (Half-integer a and Dehn-filling compatibility). Since $a = \frac{1}{2} \notin \mathbb{Z}$, the row-shearing matrix T_B in Section 3 has a half-integer entry and is *not* in $\text{GL}(4, \mathbb{Z})$. Consequently, the “subtract a , add $2b$ ” charge-lattice prescription is **not Dehn-filling compatible**: Dehn filling along a slope (P, Q) requires all basis changes to be integral, so slopes with Q odd would give a non-integer shift $a \cdot Q = Q/2 \notin \mathbb{Z}$. Only slopes satisfying $Q \in 2\mathbb{Z}$ are directly compatible.

5 Explicit Verification

We verify (1) term by term for several (m, e) values, using `q_order_half` = 12. The function $f = \eta^{\mu(m, e)} \mathcal{I}(m, e) = \eta^{(e-m)/2} \mathcal{I}(m, e)$ is checked to be palindromic at each q -order. Equivalently, $f = \eta^{-c(m, e)} \mathcal{I}(m, e)$, so the “shifted series” tables below subtract $c(m, e) = (m - e)/2$ from all η -exponents.

5.1 Pure meridian: $(m, e) = (+2, 0)$

$\mu(2, 0) = -1$; centre $c = +1$. Shift (doubled): $2c = 2$.

Table 1: Shifted series $\tilde{\mathcal{I}}(+2, 0)$ at each q -power.

q -power	Shifted η -distribution	Palindrome?
q^3	$\eta^0 : +1$	✓
q^4	$\eta^0 : +2$	✓
q^5	$\eta^{-1} : +1, \quad \eta^0 : +3, \quad \eta^1 : +1$	✓
q^6	$\eta^0 : +2$	✓
q^7	$\eta^{-1} : -1, \quad \eta^0 : -1, \quad \eta^1 : -1$	✓

Every q -level contributes a palindrome in η .

5.2 Pure longitude: $(m, e) = (0, -1)$

Centre: $c = 0 - \frac{1}{2} \cdot (-1) = +\frac{1}{2}$. Shift (doubled): $2c = 1$.

Note that the coefficients at q^6 satisfy $-1 = -1$ and $+9 = +9$ under $\eta^k \leftrightarrow \eta^{-k}$. ✓

5.3 Half-integer meridian: $(m, e) = (+1, 0)$

Centre: $c = \frac{1}{2}$. Shift (doubled): $2c = 1$. The index lives at *half-integer* q -powers (consistent with the half-integer meridian charge).

Table 2: Shifted series $\tilde{\mathcal{I}}(0, -1)$ at each q -power.

q -power	Shifted η -distribution				Palindrome?
q^1	$\eta^{-1/2} : -1,$	$\eta^{+1/2} : -1$			✓
q^2	$\eta^{-1/2} : -1,$	$\eta^{+1/2} : -1$			✓
q^3	$\eta^{-1/2} : +1,$	$\eta^{+1/2} : +1$			✓
q^4	$\eta^{-1/2} : +4,$	$\eta^{+1/2} : +4$			✓
q^5	$\eta^{-1/2} : +8,$	$\eta^{+1/2} : +8$			✓
q^6	$\eta^{-3/2} : -1,$	$\eta^{-1/2} : +9,$	$\eta^{+1/2} : +9,$	$\eta^{+3/2} : -1$	✓
q^7	$\eta^{-3/2} : -2,$	$\eta^{-1/2} : +7,$	$\eta^{+1/2} : +7,$	$\eta^{+3/2} : -2$	✓

 Table 3: Shifted series $\tilde{\mathcal{I}}(+1, 0)$ at each q -power.

q -power	Shifted η -distribution				Palindrome?
$q^{3/2}$	$\eta^{-1/2} : -1,$	$\eta^{+1/2} : -1$			✓
$q^{7/2}$	$\eta^{-1/2} : +2,$	$\eta^{+1/2} : +2$			✓
$q^{9/2}$	$\eta^{-1/2} : +5,$	$\eta^{+1/2} : +5$			✓
$q^{11/2}$	$\eta^{-1/2} : +7,$	$\eta^{+1/2} : +7$			✓
$q^{13/2}$	$\eta^{-3/2} : -1,$	$\eta^{-1/2} : +6,$	$\eta^{+1/2} : +6,$	$\eta^{+3/2} : -1$	✓

5.4 Negative meridian: $(m, e) = (-2, 0)$

Centre: $c = -1$. Shift (doubled): -2 . Each q -level contributes a single term η^0 , which is trivially palindromic. ✓

5.5 Remark on truncation artefacts

For $(m, e) = (0, +1)$ with centre $c = -\frac{1}{2}$ and shift $2c = -1$, the shifted series is palindromic at *every* q -power below q^6 :

$$q^1 : \eta^{-1/2} : -1, \eta^{+1/2} : -1 \quad \checkmark \quad \dots \quad q^5 : \eta^{-1/2} : +8, \eta^{+1/2} : +8 \quad \checkmark.$$

At q^6 the shifted distribution is $\{\eta^{-3/2} : -1, \eta^{-1/2} : +9, \eta^{+1/2} : +5, \eta^{3/2} : -1\}$, which is *not* palindromic. This is a **truncation artefact**: the summation over e_{int} is cut off at a finite range and the large- e_{int} contributions to the $\eta^{+1/2}$ channel (coefficient should match +9) are missing. The coefficient +9 that does appear in the $\eta^{-1/2}$ channel comes from smaller- e_{int} terms already fully resolved. The complete (infinite) series would be palindromic.

6 The a, b Parameters and "Subtract/Add" Prescription

6.1 Summary of parameters

Parameter	Value
a (longitude coefficient in $\mu = bm + ae$)	$+\frac{1}{2}$ (half-integer)
b (meridian coefficient in $\mu = bm + ae$)	$-\frac{1}{2}$ (half-integer)
Weyl multiplier	$\mu(m, e) = \frac{e-m}{2}$
Weyl centre	$c(m, e) = \frac{m-e}{2}$
Palindrome shift (doubled, $2c$)	$m - e \in \mathbb{Z}$

6.2 Geometric interpretation

The prescription “*subtract a units of hard edge from meridian, add $2b$ units of hard edge to longitude*” with Convention A ($a = -1, b = +\frac{1}{2}$) gives the integer matrix $T_A \in \text{GL}(4, \mathbb{Z})$, a valid symplectic charge-lattice transformation. Convention B ($a = +\frac{1}{2}, b = -\frac{1}{2}$) gives the palindrome form $f(\eta) = \eta^{bm+ae}I$ directly, but the corresponding T_B has a half-integer entry and is *not* a lattice transformation, hence not Dehn-filling compatible.

The *operationally meaningful* version (in either convention) is an affine shift of the summation variable:

$$e_{\text{int}} \longrightarrow e'_{\text{int}} = e_{\text{int}} - c(m, e) = e_{\text{int}} - \frac{m-e}{2}.$$

In the shifted variable:

$$\mathcal{I}(m, e) = \eta^{c(m, e)} \cdot \sum_{e'_{\text{int}}} \eta^{e'_{\text{int}}} \prod_{\Delta} I_{\Delta}(z_{\Delta}(e'_{\text{int}})),$$

and the η -series in brackets is palindromic by the $e'_{\text{int}} \rightarrow -e'_{\text{int}}$ symmetry (using $I_{\Delta}(-x, -p) = I_{\Delta}(x, p)$).

6.3 Why $a = +1/2$ and $b = -1/2$

The Weyl multiplier $\mu(m, e) = bm + ae$ is minus the centre: $\mu = -c = -(m - e)/2 = (e - m)/2$. Matching $\mu = bm + ae$ to $(e - m)/2 = (-1/2)m + (1/2)e$ gives

$$b = -\frac{1}{2}, \quad a = +\frac{1}{2}.$$

This is confirmed numerically from the centres of the leading- q distribution: $c(2, 0) = +1 \Rightarrow \mu(2, 0) = -1 = b \cdot 2 \Rightarrow b = -\frac{1}{2}$; $c(0, -1) = +\frac{1}{2} \Rightarrow \mu(0, -1) = -\frac{1}{2} = a \cdot (-1) \Rightarrow a = +\frac{1}{2}$. ✓

The numerical checks above confirm that the full product $\prod_{\Delta} I_{\Delta}(z_{\Delta}(e_{\text{int}}))$ satisfies the palindrome property after extracting $\eta^{c(m, e)}$, so in the convention $f(\eta) = \eta^{bm+ae}I(m, e)$:

$$\boxed{a = +\frac{1}{2}, \quad b = -\frac{1}{2}.}$$

7 Full Refined Index at Selected Values

For reference we list the raw series (before shifting) at small charges. Keys are (q -power, 2η -exp); coefficients are integers.

7.1 $(m, e) = (+2, 0)$: pure meridian

$$\mathcal{I}(+2, 0) = q^3\eta^1 + 2q^4\eta + q^5(\eta^0 + 3\eta + \eta^2) + 2q^6\eta + \dots$$

$\mu(2, 0) = -1$; multiplying by η^{-1} gives a palindrome at every q -order. ✓

7.2 $(m, e) = (0, -1)$: pure longitude

$$\mathcal{I}(0, -1) = -q(\eta^0 + \eta) - q^2(\eta^0 + \eta) + q^3(\eta^0 + \eta) + 4q^4(\eta^0 + \eta) + 8q^5(\eta^0 + \eta) + \dots$$

$\mu(0, -1) = +\frac{1}{2}$; multiplying by $\eta^{1/2}$ gives half-integer exponents, palindromic. ✓

7.3 $(m, e) = (+1, 0)$: half-integer meridian

$$\mathcal{I}(+1, 0) = -q^{3/2}(\eta^0 + \eta) + 2q^{7/2}(\eta^0 + \eta) + 5q^{9/2}(\eta^0 + \eta) + 7q^{11/2}(\eta^0 + \eta) + \dots$$

$\mu(1, 0) = -\frac{1}{2}$; multiplying by $\eta^{-1/2}$ gives a palindrome. ✓

8 Conclusion

We have verified numerically and analytically that the refined 3D index of $m004$ (figure-eight knot) satisfies the Weyl symmetry

$$f(\eta) := \eta^{bm+ae} \mathcal{I}(m, e) = f(\eta^{-1}),$$

or equivalently $\mathcal{I}(m, e) = \eta^{(m-e)/2} \cdot f(\eta)$ with f palindromic, with exact parameters (in the convention $f = \eta^{bm+ae} I$)

$$\boxed{a = +\frac{1}{2}, \quad b = -\frac{1}{2}.}$$

The parameters are extracted from the leading- q centres of the pure meridian pair $(\pm 2, 0)$ and pure longitude pair $(0, \mp 1)$, and validated (palindrome-check) up to the q^7 truncation order.

Dehn-filling note. Since $a = \frac{1}{2} \notin \mathbb{Z}$, the “subtract a , add $2b$ ” row prescription produces a matrix $T_B \notin \text{GL}(4, \mathbb{Z})$, which is *not a valid charge-lattice transformation*. Consequently Dehn filling along a slope (P, Q) requires $Q \in 2\mathbb{Z}$ for the shift $a \cdot Q = Q/2$ to be integral; slopes with odd Q are not directly compatible with this Weyl symmetry.